

Jay Vankar

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EDUCATION

- National Institute of Technology, Surat (SVNIT)** Surat, India
 - Bachelor of Technology - Artificial Intelligence; CGPA: 7.5* 2024 - 2028 (Expected)
 - Core Courses: Data Structures & Algorithms, Machine Learning Basics, Operating Systems, Automata, Computer Organization*

SKILLS SUMMARY

- Languages:** Python (Research-grade), C++ (Competitive Programming), SQL
- AI Architecture:** Attention Mechanisms, Patient Baseline Encoding, RAG Pipelines, Vector DBs
- Domain Expertise:** Longitudinal ECG Analysis, Cardiac Drift Detection, Signal Processing
- Datasets:** MIMIC-IV, Nature-cited Longitudinal ECG Cohorts, Shaoxing Hospital
- Frameworks:** PyTorch, Scikit-learn, Django, SQLite, FAISS, NumPy, Pandas

RESEARCH & PROJECTS

- Personalized Cardiac Drift Detection using Attention-Based Baseline Encoding** 2025
 - Research Project*
 - Explored personalized cardiac drift detection using longitudinal 12-lead ECG data and patient-specific baseline representations.
 - Developed an attention-based baseline encoding model in PyTorch for learning individualized cardiac features from historical ECG recordings.
 - Built an end-to-end preprocessing, training, and evaluation pipeline using MIMIC-IV and longitudinal ECG datasets.
 - Performed extensive experimentation with multiple model variants, analyzing performance using validation metrics and identifying limitations of the proposed approach.
- FloatChat - AI Conversational Interface (SIH 2025)** 2025
 - Project Lead (Independently selected topic)*
 - Architecture:** Designed an LLM-based Retrieval-Augmented Generation (RAG) system for NetCDF data ingestion and conversational discovery of oceanographic datasets.
 - Achievement:** Qualified for Internal Smart India Hackathon (SIH) 2025, ranking in the Top 50 teams at SVNIT Surat.
- Algorithmic Optimization** 2024 - Present
 - Competitive Programming & Optimization*
 - Development:** Developing time-efficient C++ solutions for dynamic programming and graph-based challenges, focusing on computational efficiency and state-space search optimization.

KEY ACHIEVEMENTS

- Initiating pioneering research in personalized cardiac monitoring leveraging the MIMIC-IV clinical database.
- Qualified Internal SIH 2025 (Top 50 teams out of total submissions at SVNIT Surat).
- Language Proficiency: English (Advanced), Hindi (Native), Gujarati (Native), Spanish (Beginner).